Reconceptualizing Shadow IT: Motivations, Consequences, and Governance in the Age of Shadow AI

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Zusammenfassung

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.
Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

Abstract

Obige Zusammenfassung in englischer Sprache.

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List of Abbreviations

SAI Shadow AI

SIT Shadow IT

SLR Structured Literature Review

1 Introduction

Alaimo and Kallinikos (2022)

2 Theoretical Foundations

2.1 Shadow IT

2.2 Motivational Theories

2.3 From Shadow IT to Shadow AI

3 Methodology

The review follows a structured approach based on the guidelines provided by Alaimo and Kallinikos (2022).

3.1 Purpose

Planning - Explicit + Comprehensive + Reproducible

Purpose of this review is to identify and analyze existing literature on the phenomenon of Shadow IT (SIT) and further the development of Shadow AI (SAI) in organizational contexts. The Structured Literature Review (SLR) aims to synthesize insights on the motivations behind SIT and SAI, their organizational consequences, and the governance approaches employed to manage these phenomena.

3.2 Draft Protocol

Planning - Explicit + Comprehensive

3.3 Practical Screening

Selection - Quantitative + Qualitative

- Content: Shadow IT in Organizations, Shadow AI in Organizations, Motivations for Shadow IT/AI, Organizational Consequences of Shadow IT/AI, Governance of Shadow IT/AI
- Language: English
- Time Frame: 2005 - present
- Publication Type: Journal Articles, Conference Proceedings, Books, Book Chapters/Sections
- Setting: Organizational/professional
- Participants: Professionals, managers, analysts, data scientists

- Artifact: Dashboards, ML interfaces, analytics tools
- Design: Empirical or strong conceptual

3.4 Literature Search

The literature search was conducted using four search strings across four databases (AISeL, EBSCO, IEEE, and Web of Science), employing advanced search with Boolean operators (`OR` and `AND`) applied to all fields or title/abstract/keywords depending on the database.

Table 1: Literature Search Strings

String	Purpose	Query
S1	Shadow IT Baseline	`"shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system"`
S2	Shadow AI Baseline	`"shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI"`
S3.1	Shadow IT × Shadow AI	`("shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system") AND ("shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI")`
S3.2	Shadow IT × AI (broader)	`"shadow IT" AND "artificial intelligence"`

String S3.2 was included to capture literature that may not explicitly use the term *Shadow AI* but discusses the intersection of Shadow IT and AI technologies.

Table 2: Database Search Results per String

String	AISeL	EBSCO	IEEE	WoS	Total
S1	296	227	39	94	656
S2	6	79	9	5	99
S3.1	4	2	0	0	6
S3.2	45	18	4	3	70
Total	**351**	**326**	**52**	**102**	**831**

Of the 255 abstracts reviewed in full, 132 were excluded for reasons including overly broad scope, lack of organisational context, or wrongly exported reference type (e.g. newspaper declared as journal article). At the full-text stage, 60 records were included, 41 were excluded, access was requested for 10 records, and 12 remained inaccessible.

Table 3: Literature Screening Process

Stage	Included	Excluded	Remaining
Records identified	–	–	831
After duplicate removal	–	124	707
Practical screening			
Language (English only)	–	23	684
Publication year (2005–present)	–	30	654
Publication type	–	15	639
Content screening			
Title screening	255	384	255
Abstract screening	123	132	123
Full-text review			
Eligible full texts	60	41	60
Access requested	–	–	10
Not accessible	–	12	–

3.5 Data Extraction

Selection - Quantitative + Qualitative

3.6 Quality Appraisal

Selection - Quantitative + Qualitative + Explicit

3.7 Study Synthesization

Execution - Quantitative + Qualitative + Quantitative & Qualitative - Explicit

3.8 Writing the Review

Execution - Explicit + Reproducible

4 Structured Literature Review Results

4.1 Overview of Identified Literature

4.2 Motivational Factors

Reports findings separately for SIT and SAI, but does not deeply interpret them.

4.3 Organizational Consequences

4.4 Governance Approaches and Countermeasures

5 Conceptual Development: Extending Shadow IT to Shadow AI

5.1 Transferable Motivations

5.2 Novel Motivations

5.3 Behavioral Shift in Knowledge Work

5.4 Integrated Conceptual Framework

6 Comparative Analysis of Shadow IT and Shadow AI

6.1 Differences in Motivational Drivers

6.2 Differences in Organizational Risk

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7 Discussion

8 Limitations and Future Research Directions

9 Conclusion

References

Alaimo, Cristina and Jannis Kallinikos (2022). "Organizations Decenterded: Data Objects, Technology and Knowledge." In: *Organization Science* 33.1, pp. 19–37.

A Appendix A